GATE
CS & DA

Artificial Intelligence

Proposition & Predictive Logic

Lecture No. 7

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Recap of Previous Lecture

Topic Predicate logic

Topic Quantifier

Topic

Topic

Topic

Turn on Slide map

Topics to be Covered

Topic Quantifier

Topic negation of a statement

Topic Predicate logic

Topic multiple Quantifier

Topic

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Predicate Logic

- Universal Quantifier:

$\forall_x P(x)$

- $P(x)$ is true for all x in domain

Predicate Logic

- Existential Quantifier:

$\exists_x P(x) \Rightarrow$ $P(x)$ is true for at least one x.

Predicate Logic

Predicate Logic

	Form	
1.	$\forall x\, P(x)$	P(x) is true for all x in
2.	$\exists x\, P(x)$	
3.	$\sim \forall x\, P(x)$	
4.	$\sim \exists x\, P(x)$	
5.	$\forall x \sim P(x)$	
6.	$\exists x \sim P(x)$	
7.	$\sim \forall x \sim P(x)$	
8.	$\sim \exists x \sim P(x)$	

Predicate Logic

	Form	
1.	$\forall x\, P(x)$	P(x) is true for all x (All are True)
2.	$\exists x\, P(x)$	P(x) is true for atleast one x (At least one True)
3.	$\sim \forall x\, P(x)$	(All are not true) P(x) is true for some x
4.	$\sim \exists x\, P(x)$	P(x) is True for no x i.e It is always false.
5.	$\forall x \sim P(x)$	for all x ~P(x) is True ⇒ for all x P(x) = false
6.	$\exists x \sim P(x)$	~P(x) = True for at least one x, P(x) = false for at least one x
7.	$\sim \forall x \sim P(x)$	P(x) is false for some x
8.	$\sim \exists x \sim P(x)$	P(x) is never false.

Negation of a statement

$1 \equiv 8 \Rightarrow \sim(\exists_x \sim P(x)) = (\forall_x P(x))$

$4 \equiv 5 \Rightarrow \sim(\exists_x P(x)) = (\forall_x \sim P(x))$

ex $\sim \exists_x (P(x) \wedge \neg Q(x)) \equiv \forall_x (\neg P(x) \vee Q(x))$

So negation of a statement

⇒ Change Quantifier

⇒ Take negation inside

$\overline{(A \wedge B)} = (\bar{A} \vee \bar{B})$

$\overline{(P \wedge Q)} = (\bar{P} \vee Q)$

Negation of a statement

$$\Rightarrow \begin{cases} \sim \forall x\,(P(x) \rightarrow Q(x)) \\ \Downarrow \\ \sim \forall x\,(\neg P(x) \vee Q(x)) \\ \Downarrow \\ \exists x\,(P(x) \wedge \neg Q(x)) \end{cases}$$

P	Q	$P \rightarrow Q$
0	0	1
0	1	1
1	0	0
1	1	1

$\nearrow \bar{P} \vee Q$

$$\overline{(A \cdot B)} \Rightarrow \bar{A} + \bar{B}$$

$$\overline{(A + B)} \Rightarrow \bar{A} \cdot \bar{B}$$

$G_l \quad G_o \quad G_l \rightarrow G_o \Rightarrow (\overline{G_l} + G_o)$

Negation of a statement

Consider the statement:

"~~Not~~ all that glitters is gold"

Predicate glitters(x) is true if x glitters and predicate gold(x) is true if x is gold. Which one of the following logical formulae represents the above statement?

(A) $\forall$: glitters(x) $\Rightarrow \neg$gold(x)

(B) $\forall$: gold(x) $\Rightarrow$ glitters(x)

(C) $\exists$x : gold(x) $\wedge \neg$ glitters(x)

(D) $\exists$x : glitters(x) $\wedge \neg$ gold(x)

All that glitter is gold

$(\forall_x \text{ glitter}(x) \longrightarrow \text{gold}(x))$

- For all x, if glitter(x) is true it implies that gold(x) will be true
- So if glitter(x) is true it implies gold(x) is true

$\sim(\forall_x \text{ glitter}(x) \longrightarrow \text{gold}(x))$

$\exists_x \sim(\overline{G_l} \vee G_o)$

$\Rightarrow \exists_x (G_l \wedge \neg G_o)$

Negation of a statement

Consider the statement:

"Not all that glitters is gold"

Predicate glitters(x) is true if x glitters and predicate gold(x) is true if x is gold. Which one of the following logical formulae represents the above statement?

(A) $\forall$: glitters(x) $\Rightarrow \neg$gold(x)

(B) $\forall$: gold(x) $\Rightarrow$ glitters(x)

(C) $\exists$x : gold(x) $\wedge \neg$ glitters(x)

(D) $\exists$x : glitters(x) $\wedge \neg$ gold(x)

Negation of a statement

Statement: "All birds can fly, except penguins."

- **Predicate:** fly(x) is true if x can fly, and bird(x) is true if x is a bird. penguin(x) is true if x is a penguin.
- **Which logical formula best represents the statement?**
  - (A) ∀x: bird(x) ⇒ fly(x)
  - (B) ∀x: bird(x) ∧ ¬penguin(x) ⇒ fly(x)
  - (C) ∃x: bird(x) ∧ ¬fly(x) → Correct
  - (D) ∀x: penguin(x) ⇒ fly(x)

for all x ⇒ Bird(x) ∧ ¬ Penguin(x) → fly

• So Bird except penguin Can fly

For atleast one x ⇒ bind(x) ∧ ¬ fly(x) = True

(x is a bind and it Cannot fly)

All bind Can fly

∀x (Bird(x) → Fly(x))

for all x, Bird(x) is true i.e x is a bird then it Can fly

(i.e if Bind(x) is true then fly(x) is also true)

Negation of a statement

Statement: "If a machine is well-maintained, it won't break down."

- **Predicate:** maintained(x) is true if machine x is well-maintained, and break_down(x) is true if machine x breaks down.
- **Which logical formula best represents the statement?**
  - (A) ∀x: maintained(x) ⇒ ¬break_down(x)
  - (B) ∃x: break_down(x) ∧ maintained(x)
  - (C) ∀x: maintained(x) ⇔ break_down(x)
  - (D) ∀x: ¬maintained(x) ⇒ break_down(x)

$\forall x$ maintained(x) $\rightarrow$ ¬breakdown(x)

Negation of a statement

Statement: "Some fruits are not sweet."

- **Predicate:** sweet(x) is true if fruit x is sweet, and fruit(x) is true if x is a fruit.
- **Which logical formula best represents the statement?**
  - (A) $\forall x$: fruit(x) $\Rightarrow$ $\neg$sweet(x)
  - (B) $\exists x$: fruit(x) $\wedge$ $\neg$sweet(x)
  - (C) $\forall x$: $\neg$fruit(x) $\Rightarrow$ sweet(x)
  - (D) $\exists x$: fruit(x) $\wedge$ sweet(x)

$\exists_x (\text{fruit}(x) \wedge \neg \text{sweet}(x))$

Negation of a statement

Statement: "Every person who studies hard passes the exam."

- **Predicate:** studies_hard(x) is true if person x studies hard, and passes_exam(x) is true if person x passes the exam.
- **Which logical formula best represents the statement?**
  - (A) ∃x: studies_hard(x) ⇒ passes_exam(x)
  - (B) ∀x: studies_hard(x) ⇒ passes_exam(x)
  - (C) ∀x: ¬studies_hard(x) ⇒ passes_exam(x)
  - (D) ∃x: passes_exam(x) ∧ ¬studies_hard(x)

Negation of a statement

Statement: "Some students are not hardworking."

$\exists x\,(S(x) \wedge \neg H(x))$

- **Predicates:**
  - $S(x)$ = x is a student.
  - $H(x)$ = x is hardworking.

What is the correct logical translation of the statement?

(A) $\forall x(S(x) \Rightarrow \neg H(x))$

(B) $\exists x(S(x) \wedge \neg H(x))$

(C) $\forall x(S(x) \wedge H(x))$

(D) $\exists x(\neg S(x) \wedge H(x))$

Negation of a statement

Statement: "Everyone who reads books is intelligent."

$\forall x (Rx \rightarrow Ix)$

- **Predicates:**
  - `R(x)` = x reads books.
  - `I(x)` = x is intelligent.

Which of the following logical formulae represents the statement?

(A) $\exists x(R(x) \Rightarrow I(x))$

(B) $\forall x(R(x) \Rightarrow I(x))$

(C) $\exists x(R(x) \wedge I(x))$

(D) $\forall x(I(x) \Rightarrow R(x))$

Negation of a statement

Statement: "No animals are capable of speech."

- **Predicates:**
  - $A(x)$ = x is an animal.
  - $S(x)$ = x is capable of speech.

(If x is animal then it cannot speak.)

$\Rightarrow \forall x\,(A(x) \rightarrow \neg S(x))$

Which of the following logical formulae represents the statement?

~~(A) $\exists x(A(x) \wedge \neg S(x))$~~

(B) $\forall x(A(x) \Rightarrow \neg S(x))$

(C) $\forall x(\neg A(x) \vee \neg S(x))$

~~(D) $\exists x(\neg A(x) \vee S(x))$~~

for all x (either x is not animal or x cannot speak)

for all x (if x is an animal then it cannot speak)

Negation of a statement

Statement: "No animals are capable of speech."

- **Predicates:**
  - $A(x)$ = x is an animal.
  - $S(x)$ = x is capable of speech.

Which of the following logical formulae represents the statement?

(A) $\exists x(A(x) \wedge \neg S(x))$

(B) $\forall x(A(x) \Rightarrow \neg S(x))$

(C) $\forall x(\neg A(x) \vee \neg S(x))$

(D) $\exists x(\neg A(x) \vee S(x))$

Negation of a statement

Statement: "There is at least one teacher who does not assign homework."

- **Predicates:**
  - $T(x)$ = x is a teacher.
  - $A(x)$ = x assigns homework.

Which of the following logical formulae represents the statement?

(A) $\exists x(T(x) \wedge \neg A(x))$

(B) $\forall x(T(x) \Rightarrow \neg A(x))$

(C) $\exists x(T(x) \Rightarrow A(x))$

(D) $\exists x(\neg T(x) \wedge A(x))$

- $\exists_x (T(x) \wedge \neg A(x))$

$A \to B \equiv \neg A \vee B$

Negation of a statement

Which of the following is the negation of $[\forall x, \alpha \to (\exists y, \beta \to (\forall u, \exists v, \gamma))]$?

(A) $[\exists x, \alpha \to (\forall y, \beta \to (\exists u, \forall v, \gamma))]$

(B) $[\exists x, \alpha \to (\forall y, \beta \to (\exists u, \forall v, \neg \gamma))]$

(C) $[\forall x, \neg \alpha \to (\exists y, \neg \beta \to (\forall u, \exists v, \neg g))]$

(D) $[\exists x, \alpha \wedge (\forall y, \beta \wedge (\exists u, \forall v, \neg g))]$

$\sim[\forall_x \alpha \to (\exists_y \beta \to (\forall_u \exists_v \gamma))]$

$\sim[\forall_x \neg\alpha \vee (\exists_y \neg\beta \vee (\forall_u \exists_v \gamma))]$

$[\exists_x \alpha \wedge \sim(\exists_y \neg\beta \vee (\forall_u \exists_v \gamma))]$

$\Leftarrow [\exists_x \alpha \wedge (\forall_y \beta \wedge \sim(\forall_u \exists_v \gamma))]$

$[\exists_x \alpha \wedge (\forall_y \beta \wedge (\exists_u \forall_v \sim\gamma))]$

Negation of a statement

Statement: "~~Not~~ all programmers are good at math."

- Predicates:
  - P(x) = x is a programmer.
  - G(x) = x is good at math.

Which of the following logical formulae represents the statement?

(A) $\exists x(P(x) \wedge \neg G(x))$

(B) $\forall x(P(x) \Rightarrow G(x))$

(C) $\exists x(P(x) \Rightarrow G(x))$

(D) $\forall x(P(x) \wedge \neg G(x))$

$\forall x\,(P(x) \rightarrow G(x))$

$\sim \forall x\,(P(x) \rightarrow G(x))$

$\sim \forall x\,(\neg P(x) \vee G(x))$

$\exists x\,(P(x) \wedge \neg G(x))$

OR $\sim \forall x\,(P(x) \rightarrow G(x))$

A≡C B≡D

Negation of a statement

Statement: "Not all programmers are good at math."

- Predicates: (A≡C) (B≡D) A
  - P(x) = x is a programmer.
  - G(x) = x is good at math.

Which of the following logical formulae represents the statement?

(A) $\exists x(P(x) \wedge \neg G(x))$

(B) $\forall x(P(x) \Rightarrow G(x))$

(C) $\exists x(P(x) \Rightarrow G(x))$

(D) $\forall x(P(x) \wedge \neg G(x))$

B

- None programmer is good at maths.

$\forall x\,(P(x) \rightarrow \neg S(x))$

- All programmer not good at maths

C D

- If x is a programmer then he is not good at maths

Negation of a statement

What is the logical translation of the following statements?

"None of my friends are perfect."

(A) $\exists x\,(F(x) \wedge \neg P(x))$

(B) $\exists x(\neg F(x) \wedge P(x))$

(C) $\exists x\,(\neg F(x) \wedge \neg P(x))$

(D) $\neg\exists x(F(x) \wedge P(x))$

[NAT]

Find negation of the statement formula

$\forall_x\{P(x) \rightarrow Q(x)\}$

done

[MCQ]

The statement formula $\forall_x\{B_{(x)} \wedge I_{(x)}\}$ is equivalent to

Predicate

Quantifier

A. $\exists_x\{B_{(x)} \rightarrow \sim I_{(x)}\}$ $\quad 1 \rightarrow 0 \Rightarrow 0$ ✓

B. $\sim \exists_x\{B_{(x)} \rightarrow \sim I_{(x)}\}$

C. $\exists_x\{\sim B_{(x)} \rightarrow I_{(x)}\}$ $\quad 0 \rightarrow 1$

D. $\sim \exists_x\{\sim B_{(x)} \rightarrow I_{(x)}\}$ $\quad 0 \rightarrow 1$

$\neg\neg \forall x\,(B(x) \wedge I(x))$

$\neg \exists x\,(\neg B(x) \vee \neg I(x))$

B	I	$\neg B \vee \neg I$
0	0	1
0	1	1
1	0	1
1	1	0

$A = \bar{\bar{A}} = \neg\neg A$

[MCQ]

P(x,y): x likes y

A. $\forall_x \forall_y P(x, y) \equiv \forall_y \forall_x P(x, y)$

B. $\exists_x \exists_y P(x, y) \equiv \exists_y \exists_x P(x, y)$

C. $\exists_x \forall_y P(x, y) \Rightarrow \forall_y \exists_x P(x, y)$

D. $\exists_y \forall_x P(x, y) \Rightarrow \forall_x \exists_y P(x, y)$

[MCQ]

Which of the following is/are true?

A. $\forall x\, \exists y\, P(x, y) \Rightarrow \exists y\, \forall x\, P(x, y)$

B. $\forall x\, \forall y\, P(x, y) \equiv \exists y\, \forall x\, P(x, y)$

C. $\exists x\, \forall y\, P(x, y) \Rightarrow \forall y\, \exists x\, P(x, y)$

D. $\exists x\, \exists y\, P(x, y) \Rightarrow \forall y\, \exists x\, P(x, y)$

[NAT]

Consider the first order logic sentence F : $\forall x(\exists y\ R(x, y))$. Assuming non-empty logical domains, which of the sentences below are implied by F?

I. $\exists y(\exists x\ R(x, y))$

II. $\exists y(\forall x\ R(x, y))$

III. $\forall y(\exists x\ R(x, y))$

IV. $\sim \exists x(\forall y \sim R(x, y))$

Predicate Logic

8. Let S(x) be the predicate "x is a multiple of 5" where x is an integer. Express the statement "There is no integer that is a multiple of 5" using predicate logic. ⇒ It means S(x) = False always ⇒ $\forall_x \sim S(x) \Rightarrow \sim \exists_x S(x)$

9. Let T(x,y) be the predicate "x is taller than y" where x and y are people. Express the statement "Nobody is taller than themselves" using predicate logic. $\forall_x \sim T(x,x) \equiv \sim \exists_x T(x,x)$.

- 10. Let P(x) be the predicate "x is a planet in our solar system" where x is a celestial body. How would you express "There are exactly 8 planets in our solar system" using predicate logic?

⇒ $T(x,x)$ ⇒ x is taller than x

So $\forall_x T(x,x)$ = False

$\forall_x \sim T(x,x)$ ✓

- $P(x_1)$ ⇒ x_1 is a planet of SS
- $P(x_2)$ = x_2 " " " " "
- ⋮
- $P(x_8)$ = x_8 - - - - - - "

Now these all shd be true together.

- (Final Predicate $\forall x_1 \forall x_2 \cdots \forall x_8 \left(P(x_1) \wedge P(x_2) \wedge P(x_3) \cdots \cdots \wedge P(x_8) \wedge \sum_{\substack{i=1 \\ j=1}}^{8} (x_i \neq x_j) \right)$)

Predicate Logic

. **Examples of Predicate Logic Statements**

- **Universal Example**:

  `∀x (Human(x) → Mortal(x))`

  Translation: "For all `x`, if `x` is a human, then `x` is mortal."

- **Existential Example**:

  `∃x (Human(x) ∧ Rich(x))`

  Translation: "There exists an `x` such that `x` is both human and rich."

- **More Complex Example**:

  `∀x ∃y (Parent(y, x) ∧ Loves(y, x))`

  Translation: "For every `x`, there exists a `y` such that `y` is a parent of `x` and `y` loves `x`."

sentences with multiple quantifiers

Quantifications of Two Variables		
Statement	***When True?***	***When False?***
$\forall x \forall y P(x, y)$ $\forall y \forall x P(x, y)$	$P(x, y)$ is true for every pair x, y.	There is a pair x, y for which $P(x, y)$ is false.
$\forall x \exists y P(x, y)$	For every x there is a y for which $P(x, y)$ is true.	There is an x such that $P(x, y)$ is false for every y.
$\exists x \forall y P(x, y)$	There is an x for which $P(x, y)$ is true for every y.	For every x there is a y for which $P(x, y)$ is false.
$\exists x \exists y P(x, y)$	There is a pair x, y for which $P(x, y)$ is true.	$P(x, y)$ is false for every pair x, y.

. $\forall x \exists y (x + y = 0)$ is read as:

a. ○ For all x and y, $x + y = 0$

b. ○ For all y, there exists an x such that $x + y = 0$

c. ○ For all x, there exists a y such that $x + y = 0$

d. ○ For all x and y, $xy = 0$

Express the statement "for every x and for every y, $x + y > 10$"

a. ○ $\forall x \forall y P(y, y)$

b. ○ $\forall x \forall y P(x, y)$

c. ○ $\forall y P(x > y)$

d. ○ $\forall x \forall y P(x)$

sentences with multiple quantifiers

sentences with multiple quantifiers

$\exists x\ \forall y\ (P(x, y)) \Rightarrow \forall y\ \exists x\ (P(x, y))$

- **Interpretation of $\exists x\ \forall y\ (P(x, y))$**: There exists an **x** such that for all **y**, the property **P(x, y)** holds. This means there is a single **x** that works for all **y**.

- **Interpretation of $\forall y\ \exists x\ (P(x, y))$**: For every **y**, there exists an **x** such that the property **P(x, y)** holds. This means that for each **y**, we can find some **x**, but the **x** could be different for different values of **y**.

sentences with multiple quantifiers

- $\forall y\ \exists x\ (P(x, y)) \Leftarrow \exists x\ \forall y\ (P(x, y))$ (reverse implication)
- However, the reverse is not true.
- $\forall y\ \exists x\ (P(x, y))$ says that for each y, we can find an x such that P(x, y) holds, but it doesn't guarantee that there is one single x that works for all y.
- In other words, different values of y might require different values of x. Therefore, $\forall y\ \exists x\ (P(x, y))$ does not imply $\exists x\ \forall y\ (P(x, y))$.

sentences with multiple quantifiers

- Let $P(x, y)$ be "$x + y = 0$".
- $\exists x \forall y P(x, y)$: There exists a single x such that for all y, $x + y = 0$. This is false because no single x satisfies the equation for all y.
- $\forall y \exists x P(x, y)$: For every y, there exists some x such that $x + y = 0$. This is true because for any y, we can choose $x = -y$, but the value of x depends on y.
- Thus, $\forall y \exists x (P(x, y))$ holds, but $\exists x \forall y (P(x, y))$ does not, showing that the reverse implication doesn't hold.

sentences with multiple quantifiers

- $\forall x \forall y P(x, y)$:

sentences with multiple quantifiers

- $\exists x \exists y P(x, y)$:

sentences with multiple quantifiers

1. ∀x ∃y (Loves(x, y))
   - "For every person x, there exists a person y such that x loves y."
2. ∃x ∀y (Loves(x, y))
   - "There exists a person x such that for every person y, x loves y."
3. ∀x ∃y (Teaches(y, x))
   - "For every student x, there exists a teacher y such that y teaches x."

sentences with multiple quantifiers

4. $\exists x\ \forall y\ (Friend(y, x))$
   - "There exists a person x who is a friend to every person y."
5. $\forall x\ \forall y\ (Person(x) \land Person(y) \rightarrow \exists z\ (Friend(x, z) \land Friend(y, z)))$
   - "For every pair of people x and y, there exists a person z who is a friend to both x and y."
6. $\forall x\ \exists y\ \forall z\ (Knows(x, z) \rightarrow Knows(y, z))$
   - "For every person x, there exists a person y such that y knows everyone that x knows."

THANK - YOU